

DEMO-003 - Submission

Mixed typeset response and annotations - synthetic student

Q1 - Calculus

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With $u=x^2$, $I=(e-1)/2$.

Q2 - Mechanics

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$a = 9.81(\sin 30 \text{ deg} - 0.2 \cos 30 \text{ deg}) = 3.20$; therefore v is approximately 4.4 m/s.

unit missing here



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Q3 - Linear algebra

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A is invertible, so multiplying by A^T cannot change the kernel. Therefore A and $A^T A$ have equal rank.

A may be rectangular - this assumption is not available.

Q4 - Programming

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```
if not xs: return []
m = max(xs)
z = [exp(x - m) for x in xs]
return [value / sum(z) for value in z]
```